

Functional Requirement Document (FRD)

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Project Title

Pharma Sales Force Effectiveness Analytics System

1. Purpose

The purpose of this document is to define the functional requirements of the Pharma Sales Force Effectiveness Analytics System.

The system will analyze pharmaceutical sales, physician, territory, and marketing data and provide dashboards, reports, and KPIs to support business decision-making.

2. Scope

The system shall provide:

- Territory Performance Analytics
- Sales Representative Performance Analytics
- Physician Analytics
- Product Performance Analytics
- Marketing ROI Analytics
- Interactive Dashboard Reporting

3. Users

User Role	Description
Sales Manager	Monitor territory and sales rep performance
Marketing Manager	Monitor marketing effectiveness
Commercial Analytics Team	Perform business analysis
Leadership Team	Strategic decision making

4. Functional Requirements

Module 1: Revenue Analytics

FR-01

The system shall calculate total revenue.

FR-02

The system shall display revenue by region.

FR-03

The system shall display revenue by product.

FR-04

The system shall display monthly revenue trends.

FR-05

The system shall identify top-performing territories.

Module 2: Sales Representative Analytics

FR-06

The system shall calculate sales achieved by each representative.

FR-07

The system shall calculate target achievement percentage.

FR-08

The system shall rank sales representatives based on performance.

FR-09

The system shall compare sales representative performance across territories.

FR-10

The system shall identify underperforming representatives.

Module 3: Physician Analytics

FR-11

The system shall calculate prescription volume by physician.

FR-12

The system shall identify high-value physicians.

FR-13

The system shall categorize physicians by specialty.

FR-14

The system shall display physician contribution to total prescriptions.

FR-15

The system shall rank physicians based on prescription volume.

Module 4: Product Analytics

FR-16

The system shall calculate revenue by medicine.

FR-17

The system shall identify top-selling medicines.

FR-18

The system shall compare product performance across regions.

FR-19

The system shall display product sales trends.

Module 5: Marketing Analytics

FR-20

The system shall calculate marketing spend by territory.

FR-21

The system shall calculate marketing ROI.

FR-22

The system shall compare marketing effectiveness across regions.

FR-23

The system shall identify territories generating highest ROI.

Module 6: Dashboard and Reporting

FR-24

The system shall provide an executive summary dashboard.

FR-25

The system shall provide KPI cards.

FR-26

The system shall support filtering by:

- Region
- Product

- Sales Representative
- Physician
- Month

FR-27

The system shall provide drill-down analysis.

FR-28

The system shall support interactive visualizations.

FR-29

The system shall provide downloadable reports.

FR-30

The system shall display business insights and trends.

5. KPI Definitions

KPI	Description
Revenue	Total sales generated
Achievement %	Sales target attainment
Prescription Volume	Total prescriptions generated
Marketing ROI	Return on marketing investment
Revenue by Territory	Territory performance
Rep Productivity	Revenue generated per representative

6. Input Data Requirements

Field	Description
Date	Transaction date
Region	Territory name

Field	Description
Sales_Rep	Representative name
Physician	Doctor name
Specialty	Physician specialty
Product	Medicine name
Revenue	Sales amount
Prescriptions	Number of prescriptions
Marketing_Spend	Campaign expenditure
Target_Sales	Sales target

7. Data Validation Rules

Rule	Validation
Revenue	Cannot be negative
Prescriptions	Cannot be negative
Marketing Spend	Cannot be negative
Achievement %	Must be between 0–200%
Null Region	Reject record

8. Business Rules

BR-01

Revenue must be greater than or equal to zero.

BR-02

Marketing Spend must be greater than or equal to zero.

BR-03

Prescription count cannot be negative.

BR-04

Achievement % shall be calculated using:

$$\text{Achievement \%} = \text{Actual Sales} / \text{Target Sales} \times 100$$

BR-05

Marketing ROI shall be calculated using:

$$\text{ROI \%} = \text{Sum(Revenue)} / \text{Sum(marketing_spend)}$$

BR-06

Top-performing territory shall be the territory with highest revenue.

BR-07

High-value physicians shall be identified based on prescription volume.

9. Report Requirements

Report 1

Territory Performance Report

Contents:

- Revenue by Territory
- Territory Ranking
- Revenue Trend

Report 2

Sales Representative Performance Report

Contents:

- Sales Achieved
- Target Achievement %
- Ranking

Report 3

Physician Analysis Report

Contents:

- Prescription Volume
- Physician Ranking
- Specialty Analysis

Report 4

Marketing ROI Report

Contents:

- Marketing Spend
- Revenue Generated
- ROI %

10. Non-Functional Requirements

Performance

Dashboard should load within 5 seconds.

Availability

System should be available during business hours.

Usability

Dashboard should be easy to navigate.

Scalability

System should support increasing data volumes.

Reliability

Reports should produce consistent results.

11. Success Criteria

- Users can analyze territory performance.
- Users can monitor sales rep effectiveness.
- Users can identify high-value physicians.
- Users can evaluate marketing ROI.
- Dashboards support data-driven decisions.
- KPI calculations are accurate and automated.